1 Dimension

The next lemma is a key result which we will use repeatedly.

Lemma 1.1 (Plus / minus lemma)**.** *Let S be a subset of a vector space V.*

(a) If S is linearly independent and v is not in the span of S, then we can add v to S and S will still be linearly independent. That is, if S is linearly independent and $v \in V - span(S)$, then $S \cup \{v\}$ is linearly independent.

(b) If $v \in S$ is in the span of the other vectors in S, then we can remove v from S without changing the span of S. That is, $span(S) = span(S - \{v\})$.

Exercise 1.2. Prove Lemma 1.1.

Recall that a *finite-dimensional vector space* is a vector space with a finite spanning set.

Theorem 1.3. *Let V be a finite-dimensional vector space and S be a subset of V. Then*

(a) If $span(S) = V$, then S contains a subset B which is a basis of V.

(b) If S is linearly independent, then S can be extended to a basis of V.

Exercise 1.4. Prove Theorem 1.3.

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Corollary 1.5. *Every finite-dimensional vector space has a finite basis.*

Exercise 1.6. Prove Corollary 1.5.

Theorem 1.7. *Let V be a finite-dimensional vector space which is spanned by a set of m vectors $\{v_1, \ldots, v_m\}$. Then any linearly independent set of vectors in V is finite and contains no more than m elements.*

Proof. To prove the theorem it suffices to show that every subset S of V which contains more than m vectors is linearly dependent. Let S be such a subset. Then S contains distinct vectors $w_1, w_2, \ldots, w_n$ where $n > m$. Since $\{v_1, \ldots, v_m\}$ spans V, we can write each w_i as a linear combination

$$w_i = \sum_{k=1}^{m} \beta_{ik} v_k. \tag{1.8}$$

Suppose there are scalars $\alpha_1, \ldots, \alpha_m \in F$ such that

$$0 = \sum_{i=1}^{n} \alpha_i w_i. \tag{1.9}$$

Substituting (1.8) into (1.9) and rearranging gives

$$\begin{aligned} 0 &= \sum_{i=1}^{n} \alpha_i \left(\sum_{k=1}^{m} \beta_{ik} v_k \right) \\ &= \sum_{i=1}^{n} \sum_{k=1}^{m} \alpha_i (\beta_{ik} v_k) \quad \text{(distributive property over vector addition)} \\ &= \sum_{i=1}^{n} \sum_{k=1}^{m} (\alpha_i \beta_{ik}) v_k \quad \text{(associative property of scalar multiplication)} \\ &= \sum_{k=1}^{m} \left(\sum_{i=1}^{n} \alpha_i \beta_{ik} \right) v_k \quad \text{(associative and commutative properties of vector addition)} \end{aligned} \tag{1.10}$$

Consider the homogeneous linear system

$$\sum_{i=1}^{n} \alpha_i \beta_{ik} = 0, \quad 1 \leq k \leq m.$$

for $k = 1, \ldots, m$. This is a system of m equations (one for each k) in $n > m$ unknowns (the α_is). We have seen that a homogeneous linear system with fewer equations than unknowns has a nontrivial solution (since the system will have free variables). This nontrivial solution for the $\alpha_i s$ satisfies (1.10) (since the coefficient of each v_k is zero and hence the sum over k is zero) and therefore satisfies (1.9) (since these equations are equivalent). This shows that (1.9) has a nontrivial solution for the α_is, this shows that S is linearly dependent. □

Corollary 1.11. *Every basis of a finite-dimensional vector space is finite.*

Exercise 1.12. Use Theorem 1.7 to prove Corollary 1.11.

Corollary 1.13. *If V is a finite-dimensional vector space, then any two bases of V have the same number of elements.*

Exercise 1.14. Prove Corollary 1.13. (*Hint*: Suppose $\{v_1, \ldots, v_m\}$ and $\{w_1, \ldots, w_n\}$ are two bases of V. Use Theorem 1.7 to prove that $m = n$.)

Definition 1.15 (Dimension)**.** Let V be a finite-dimensional vector space. The number of elements in any basis of V is called the *dimension* of V, and is denoted by $\dim V$.

Example 1.16.

(a) $\emptyset$ is a basis of $\{0\}$, so $\dim\{0\} = 0$.

(b) The set $\{e_i\}_{i=1}^n$ (where $e_i \in F^n$ has a 1 in the ith entry and 0s elsewhere) is a basis of F^n, so $\dim F^n = n$.

Corollary 1.17. *Let V be a finite-dimensional vector space and let $n = \dim V$. Then*

(a) any subset of V which contains more than n vectors is linearly dependent;

(b) no subset of V which contains fewer than n vectors can span V.

Exercise 1.18. Prove Corollary 1.17. (*Hint*: Part (a) follows from Theorem 1.7. Part (b) follows from Corollary 1.13 and Theorem 1.3.

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Corollary 1.19. *Let V be a finite-dimensional vector space of dimension* $\dim V = n$*. Then*

(a) Any set of n linearly independent vectors of V is a basis of V.

(b) Any set of n vectors which spans V is a basis of V.

Exercise 1.20. Prove Corollary 1.19. (*Hint*: Use Theorem 1.3.)

Example 1.21. Consider the set $S = \left\{ \begin{bmatrix} -5 \\ 3 \end{bmatrix}, \begin{bmatrix} 21 \\ 16 \end{bmatrix} \right\} \subset \mathbb{R}^2$. Since these two vectors are not scalar multiples of each other, S is linearly independent and hence a basis of $\mathbb{R}^2$ by Theorem 1.19.